



FINAL MARK

**GIRRAWEEEN HIGH SCHOOL
MATHEMATICS EXTENSION 2
HSC ASSESSMENT TASK 1
ANSWERS COVER SHEET**

Name: _____

QUESTION	MARK	E2	E3	E4	E5	E6	E7	E8	E9
1-5	/5		✓						✓
6	/16		✓						✓
7	/18		✓						✓
8	/17		✓						✓
9	/16		✓						✓
10	/13		✓						✓
11	/13		✓						✓
TOTAL									
	/98		/98						/98

HSC Outcomes

Mathematics Extension 2

- E2 chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings.
- E3 uses the relationship between algebraic and geometric representations of complex numbers and of conic sections.
- E4 uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials.
- E5 uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces, resisted motion and circular motion
- E6 combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions.
- E7 uses the techniques of slicing and cylindrical shells to determine volumes.
- E8 applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems.
- E9 communicates abstract ideas and relationships using appropriate notation and logical argument.



GIRRAWEE HIGH SCHOOL

HSC Task 1

YEAR 11 Mathematics Extension 2 2013

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in Questions 6 - 10. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- For Questions 1 - 5, write the letter corresponding to the correct answer in your answer booklet. For Questions 6 – 10, each question is to be returned on a *separate* piece of paper clearly marked Question 6, Question 7, etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.

Multiple Choice(5 marks)

Write the letter corresponding to the correct answer in your answer booklet.

1 Let $z = 1 + i$ and $w = 1 - 2i$. What is the value of zw ?

- (A) $-1 - i$
- (B) $-1 + i$
- (C) $3 - i$
- (D) $3 + i$

2 Let $z = 3 - 4i$ and $w = \sqrt{3} + i$. What is the value of $\frac{z}{w}$?

- (A) $\frac{3\sqrt{3} + 4}{4} + \frac{(-4\sqrt{3} - 3)i}{4}$
- (B) $\frac{3\sqrt{3} - 4}{4} + \frac{(-4\sqrt{3} - 3)i}{4}$
- (C) $\frac{3\sqrt{3} + 4}{2} + \frac{(-4\sqrt{3} - 3)i}{2}$
- (D) $\frac{3\sqrt{3} - 4}{2} + \frac{(-4\sqrt{3} - 3)i}{2}$

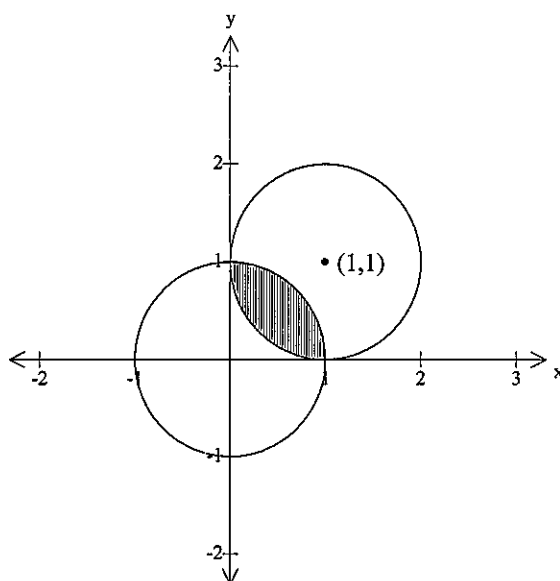
3 Let $z = 3 - i$. What is the value of $\overline{iz}$?

- (A) $-1 - 3i$
- (B) $-1 + 3i$
- (C) $1 - 3i$
- (D) $1 + 3i$

4 What is $-2 + 2\sqrt{3}i$ expressed in modulus-argument form?

- (A) $2(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$
- (B) $4(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$
- (C) $2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$ The
- (D) $4(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$

5 Consider the Argand diagram below.



Which inequality could define the shaded area?

- (A) $|z| \leq 1$ and $|z - (1 - i)| \geq 1$
- (B) $|z| \leq 1$ and $|z - (1 + i)| \geq 1$
- (C) $|z| \leq 1$ and $|z - (1 - i)| \leq 1$
- (D) $|z| \leq 1$ and $|z - (1 + i)| \leq 1$

Question 6(16 marks)

- a. (i) Express $\sqrt{3} + i$ and $1 - i$ in modulus-argument form. 4
- (ii) Express $\frac{(\sqrt{3} + i)^6}{(1 - i)^8}$ in the form $x + iy$. 3
- b. Find the five fifth roots of $1 + \sqrt{3}i$. Show these on an Argand Diagram and find the area of the pentagon formed by the five points representing these roots. 5
- c. Let $w = \frac{3 + 4i}{5}$ and $z = \frac{5 + 12i}{13}$,
- (i) Find wz and $\overline{wz}$ in the form $x + iy$. 2
- (ii) Hence, find two distinct ways of writing 65^2 as a sum of $a^2 + b^2$, where a and b are integers and $0 < a < b$ and $|w| = |z| = 1$. 2

Question 7(18 marks)

- a. (i) Find all real numbers x and y such that $(x + iy)^2 = -3 + 4i$ 4
- (ii) Hence solve $z^2 - 3z + (3 - i) = 0$ 3
- b. By using De Moivre's theorem and the Binomial theorem, obtain an expression for
- (i) $\sin 4\theta$ 2
- (ii) $\cos 4\theta$ 2
- (iii) $\tan 4\theta$ 2
- c. Given that ω is a complex root of $z^3 = 1$,
- (i) Show that ω^2 is also a root of the equation. 1
- (ii) Show that $1 + \omega + \omega^2 = 0$. 1
- (iii) Show that $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^5)(1 - \omega^7)(1 - \omega^8) = 27$ 3

Question 8(17 marks)

- a. (i) Show that if $z = \cos \theta + i \sin \theta$ then $z^n + \frac{1}{z^n} = 2 \cos n\theta$. 3
- (ii) Hence express $\cos^5 \theta$ in terms of $\cos 5\theta, \cos 3\theta, \cos \theta$. 4
- b. (i) Find the six sixth roots of unity. Write your answer in mod-arg form. 4
- (ii) Resolve $z^6 - 1$ into real quadratic factors. 3
- (iii) Hence show that $\cos \frac{\pi}{3} \cos \frac{2\pi}{3} = -\frac{1}{4}$ 3

Question 9(16 marks)

- a. Sketch the following on separate Argand Diagrams. Find the Cartesian equation and describe the locus.

(i) $\arg(z+2) = \frac{\pi}{4}$ (ii) $|z+2-i| = 4$ (iii) $|z-2i| = |z-4|$ 9

- b. (i) On the same diagram, draw a neat sketch of the locus described by:

I. $|z-(3+2i)| = 2$ 4

II. $|z+3| = |z-5|$

- (ii) Hence, write down all the values of z which satisfy simultaneously 1

$|z-(3+2i)| = 2$ and $|z+3| = |z-5|$

- (iii) Use your diagram in (i) to determine the values of k for which the simultaneous equations for $|z-(3+2i)| = 2$ and $|z-2i| = k$ have exactly one solution for z . 2

Question 10(13 marks)

- a. Let $z = a+ib$ where $a^2 + b^2 \neq 0$

(i) Show that if $\text{Im}(z) > 0$, then $\text{Im}\left(\frac{1}{z}\right) < 0$. 4

(ii) Prove that $\left|\frac{1}{z}\right| = \frac{1}{|z|}$ 3

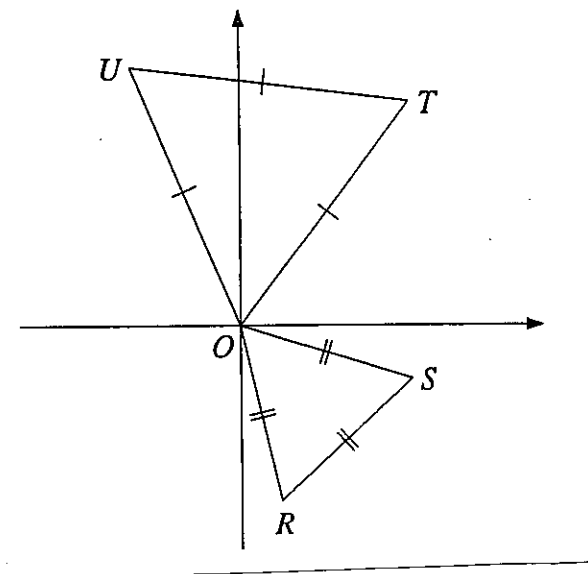
b. Show that the roots of $(z-1)^4 + (z+1)^4 = 0$ are $\pm i \cot\left(\frac{\pi}{8}\right)$ and $\pm i \cot\left(\frac{3\pi}{8}\right)$. 6

Question 11(13 marks)

a. Sketch $\arg\left(\frac{z-3}{z+1}\right) = \frac{\pi}{3}$. Describe the locus and find its Cartesian equation.

6

b.



The diagram shows points O , R , S , T and U in the complex plane. These points correspond to the complex numbers O , r , s , t and u respectively. The triangles ORS and OTU are equilateral.

Let $\omega = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$.

(i) Explain why $u = \omega t$.

1

(ii) Find the complex number r in terms of s and $\bar{\omega}$.

2

(iii) Using the complex numbers, show that the lengths of RT and SU are equal.

4

END OF EXAMINATION

HSC Task 1 Mathematics Extension 2

SOLUTIONS 2013

Multiple Choice

1. C 2. B 3. C 4. B 5. D

Question 6 (16 marks)

9) (i) $\sqrt{3} + i = 2 \operatorname{cis} \frac{\pi}{6}$ (2)

$1 - i = \sqrt{2} \operatorname{cis} \frac{-\pi}{4}$ (2)

$$\begin{aligned} \text{ii) } \frac{(\sqrt{3} + i)^6}{(1 - i)^8} &= \frac{(2 \operatorname{cis} \frac{\pi}{6})^6}{(\sqrt{2} \operatorname{cis} \frac{-\pi}{4})^8} \\ &= \frac{64 \operatorname{cis} \pi}{16 \operatorname{cis} -2\pi} \\ &= \frac{4(\cos \pi + i \sin \pi)}{\cos(-2\pi) + i \sin(-2\pi)} \\ &= -4 \end{aligned} \quad (3)$$

b) Let $z^5 = 1 + \sqrt{3}i$
 $= 2 \operatorname{cis} \frac{\pi}{3} = 2 \operatorname{cis} (\frac{\pi}{3} + 2k\pi)$
 $\therefore z = 2^{\frac{1}{5}} \left(\frac{\pi}{3} + 2k\pi \right), k=0,1,\dots,4$

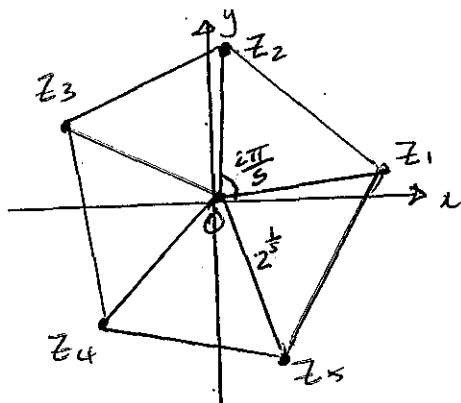
$$z_1 = 2^{\frac{1}{5}} \operatorname{cis} \frac{\pi}{15}$$

$$z_2 = 2^{\frac{1}{5}} \operatorname{cis} \frac{7\pi}{15}$$

$$z_3 = 2^{\frac{1}{5}} \operatorname{cis} \frac{13\pi}{15}$$

$$z_4 = 2^{\frac{1}{5}} \operatorname{cis} \frac{19\pi}{15} = 2^{\frac{1}{5}} \operatorname{cis} \frac{-11\pi}{15}$$

$$z_5 = 2^{\frac{1}{5}} \operatorname{cis} \frac{25\pi}{15} = 2^{\frac{1}{5}} \operatorname{cis} \frac{-\pi}{3}$$



$$\text{Area} = 5 \times \frac{1}{2} \times 2^{\frac{1}{5}} \times 2^{\frac{1}{5}} \sin \frac{2\pi}{5}$$

$$= 3.14 \text{ sq. units}$$

(5)

c) $w = \frac{3+4i}{5}; z = \frac{5+12i}{13}$

$$\begin{aligned} \text{i) } wz &= \frac{(3+4i)(5+12i)}{65} \\ &= \frac{15+36i+20i-48}{65} \\ &= \frac{-33}{65} + \frac{56i}{65} \end{aligned} \quad (1)$$

$$\begin{aligned} \text{ii) } w\bar{z} &= \frac{(3+4i)(5-12i)}{65} \\ &= \frac{63}{65} - \frac{16i}{65} \end{aligned} \quad (1)$$

$$\begin{aligned} \text{iii) } |wz| &= |w||z| = \sqrt{\left(\frac{33}{65}\right)^2 + \left(\frac{56}{65}\right)^2} \\ |w| &= |z| = 1 \\ \therefore \left(\frac{33}{65}\right)^2 + \left(\frac{56}{65}\right)^2 &= 1 \\ \therefore 65^2 &= 33^2 + 56^2 \end{aligned} \quad (1)$$

$$\begin{aligned} \text{Also } |w\bar{z}| &= |w||\bar{z}| = \sqrt{\left(\frac{63}{65}\right)^2 + \left(\frac{16}{65}\right)^2} \\ |w| &= |\bar{z}| = 1 \quad \& \quad |\bar{z}| = |z| \\ \therefore \left(\frac{63}{65}\right)^2 + \left(\frac{16}{65}\right)^2 &= 1 \\ \therefore 65^2 &= 63^2 + 16^2 \end{aligned} \quad (1)$$

Question 7 (18 marks)

a) i) $(x+iy)^2 = -3+4i$

$$x^2 + 2xyi - y^2 = -3 + 4i$$

Equating real & imaginary parts,

$$x^2 - y^2 = -3 \text{ --- (1)}; \quad 2xy = 4$$

$$y = \frac{2}{x} \text{ --- (2)}$$

Substitute (2) into (1)

$$x^2 - \left(\frac{2}{x}\right)^2 + 3 = 0$$

$$x^2 - \frac{4}{x^2} + 3 = 0$$

$$x^4 + 3x^2 - 4 = 0$$

$$(x^2 + 4)(x^2 - 1) = 0$$

no real solutions $x^2 - 1 = 0$
 $x = \pm 1$

when $x = 1, y = 2$

$x = -1, y = -2$

ii) $z^2 - 3z + (3-i) = 0$

$$z = \frac{3 \pm \sqrt{9 - 4(1)(3-i)}}{2}$$

$$= \frac{3 \pm \sqrt{-3 + 4i}}{2}$$

$$= \frac{3 \pm (1 + 2i)}{2}$$

$$z = 2 + i, 1 - i$$

b) $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$ --- (1)
(de Moivre's Thm)

$$(\cos \theta + i \sin \theta)^4 = \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta$$
 --- (2)

Equating real and imaginary parts in (1) and (2)

i) $\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta$

ii) $\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$

iii) $\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$

$$= \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta}$$

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

(dividing numerator & denominator by $\cos^4 \theta$)

c) w is a solution of $z^3 = 1$ --- (6)

i) If w is a solution, then

$$w^3 = 1$$

$$(w^2)^3 = (w^3)^2 = 1^2 = 1$$

$\therefore w^2$ is also a root. --- (1)

ii) roots of $z^3 = 1$ are $1, w \neq w^2$.

Sum of the roots $= -\frac{b}{a} = 0$

$$\therefore 1 + w + w^2 = 0$$

iii) $(1-w)(1-w^2)(1-w^4)(1-w^5)(1-w^7)(1-w^8)$

$$= (1-w)(1-w^2)(1-w)(1-w^2)(1-w)(1-w^2)$$

$$= (1-w)^3 (1-w^2)^3$$

$$= [(1-w)(1-w^2)]^3$$

$$= (1 - w^2 - w + w^3)^3$$

$$= (1 - w^2 - w + 1)^3$$

$$= (2 - (w + w^2))^3$$

$$= (2 - (-1))^3$$

$$= 27$$

(3)

Question 8 (17 marks)

a) i) $z = \cos \theta + i \sin \theta$

$$z^n = \cos n\theta + i \sin n\theta$$

$$\frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$\begin{aligned} z^n + \frac{1}{z^n} &= \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ &= 2 \cos n\theta \end{aligned} \quad (3)$$

ii) $z^5 + \frac{1}{z^5} = 2 \cos 5\theta$

$$\left(z + \frac{1}{z}\right)^5 = z^5 + 5z^3 + 10z + \frac{10}{z} + \frac{5}{z^3} + \frac{1}{z^5}$$

$$(2 \cos \theta)^5 = \left(z^5 + \frac{1}{z^5}\right) + 5\left(z^3 + \frac{1}{z^3}\right) + 10\left(z + \frac{1}{z}\right)$$

$$32 \cos^5 \theta = 2 \cos 5\theta + 10 \cos 3\theta + 20 \cos \theta$$

$$\cos^5 \theta = \frac{1}{16} \cos 5\theta + \frac{5}{16} \cos 3\theta + \frac{5}{8} \cos \theta$$

$$\cos^5 \theta = \frac{1}{16} (\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta) \quad (4)$$

b) i) $z^6 = 1 = \text{cis } 0$

$$= \text{cis } (0 + 2k\pi)$$

$$z = \text{cis } \left(\frac{2k\pi}{6}\right)$$

$$= \text{cis } \left(\frac{k\pi}{3}\right) \quad k=0, 1, \dots, 5$$

$$z_1 = \text{cis } 0 = 1$$

$$z_2 = \text{cis } \frac{\pi}{3}$$

$$z_3 = \text{cis } \frac{2\pi}{3}$$

$$z_4 = \text{cis } \pi = -1$$

$$z_5 = \text{cis } \frac{4\pi}{3} = \text{cis } \frac{-2\pi}{3}$$

$$z_6 = \text{cis } \frac{5\pi}{3} = \text{cis } \frac{-\pi}{3} \quad (4)$$

ii) $z^6 - 1 = (z-1)(z+1)(z - \text{cis } \frac{\pi}{3})(z - \text{cis } \frac{-\pi}{3})(z - \text{cis } \frac{2\pi}{3})(z - \text{cis } \frac{-2\pi}{3})$

$$z^6 - 1 = (z^2 - 1)(z^2 - 2z \cos \frac{\pi}{3} + 1)(z^2 - 2z \cos \frac{2\pi}{3} + 1) \quad (3)$$

iii)
$$\begin{array}{r} z^4 + z^2 + 1 \\ z^2 - 1 \overline{) z^6 + 0z^5 + 0z^4 + 0z^3 + 0z^2 + 0z - 1} \\ \underline{z^6 - 0z^5 - z^4} \\ z^4 + 0z^3 + 0z^2 \\ \underline{z^4 + 0z^3 - z^2} \\ z^2 + 0z - 1 \end{array}$$

(OR) $z^6 - 1 = (z^2)^3 - 1$

$$= (z^2 - 1)(z^4 + z^2 + 1)$$

$$\begin{array}{r} z^2 + 0z - 1 \\ \underline{z^2 + 0z - 1} \\ 0 \end{array}$$

$$z^6 - 1 = (z^2 - 1)(z^4 + z^2 + 1) \quad (2)$$

From (1) & (2)

$$(z^2 - 1)(z^4 + z^2 + 1) = (z^2 - 1)(z^2 - 2z \cos \frac{\pi}{3} + 1)(z^2 - 2z \cos \frac{2\pi}{3} + 1)$$

$$\therefore z^4 + z^2 + 1 = (z^2 - 2z \cos \frac{\pi}{3} + 1)(z^2 - 2z \cos \frac{2\pi}{3} + 1)$$

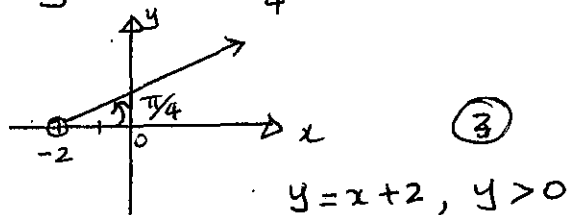
Equating coefficients of z^2

$$2 + 4 \cos \frac{\pi}{3} \cos \frac{2\pi}{3} = 1$$

$$\cos \frac{\pi}{3} \cos \frac{2\pi}{3} = -\frac{1}{4} \quad (3)$$

Question 9 (16 marks)

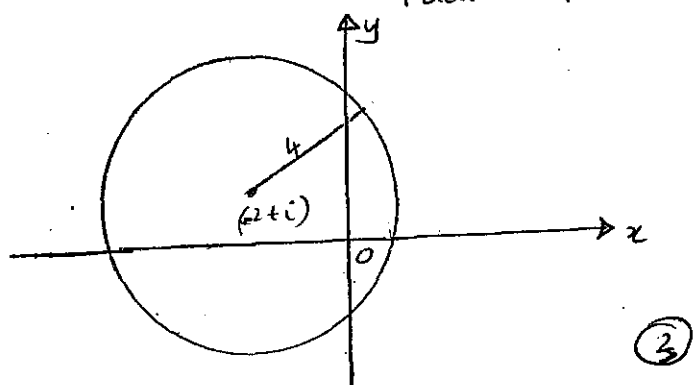
a) i) $\arg(z+2) = \frac{\pi}{4}$



ii) $|z + 2 - i| = 4$

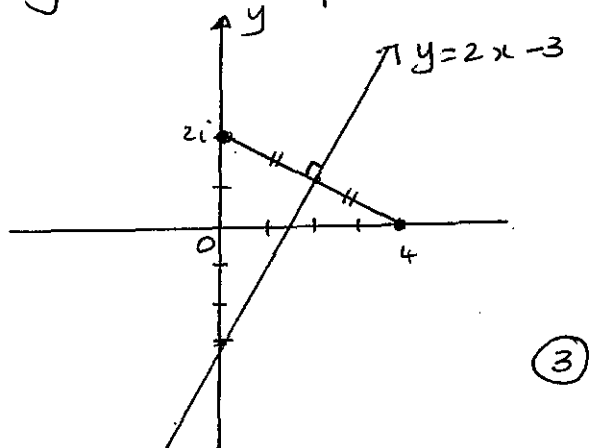
$|z - (-2 + i)| = 4$

circle, centre $(-2, 1)$
radius 4



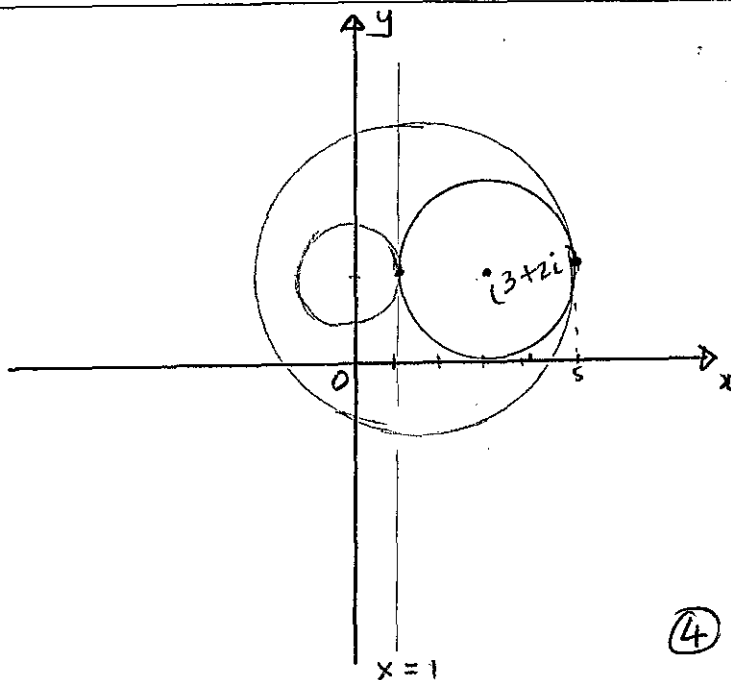
iii) $|z - 2i| = |z - 4|$

locus is the perpendicular bisector of the line segment joining $2i$ and 4 .



b) i) I. $|z - (3 + 2i)| = 2$ is a circle centre $(3 + 2i), r = 2$

II. $|z + 3| = |z - 5|$ is a perpendicular bisector of interval joining -3 & 5
ie line $x = 1$



ii) The line $x = 1$ is a tangent to the circle.

$\therefore z = 1 + 2i$ is the only point of intersection. (1)

iii) $|z - 2i| = k$ represents a circle, centre $2i$, radius k .

Hence the equations

$|z - (3 + 2i)| = 2$ and

$|z - 2i| = k$ have exactly one solution when

$k = 1$ or $k = 5$.

(see diagram in part(i)) (2)

Question 10 (13 marks)

a) Let $z = a + ib$ where $a^2 + b^2 \neq 0$

$$\begin{aligned} \text{i) } \frac{1}{z} &= \frac{1}{a+ib} \\ &= \frac{a-ib}{(a+ib)(a-ib)} \\ &= \frac{a-ib}{a^2+b^2} \end{aligned}$$

$$\operatorname{Im}\left(\frac{1}{z}\right) = \frac{-b}{a^2+b^2}$$

If $\operatorname{Im}(z) > 0$, i.e. $b > 0$,

$$\text{then } \operatorname{Im}\left(\frac{1}{z}\right) = \frac{-b}{a^2+b^2} < 0$$

(since $a^2+b^2 > 0$)

(4)

$$\text{ii) } \left| \frac{1}{z} \right| = \left| \frac{a-ib}{a^2+b^2} \right|$$

$$= \sqrt{\left(\frac{a}{a^2+b^2}\right)^2 + \left(\frac{-b}{a^2+b^2}\right)^2}$$

$$= \sqrt{\frac{a^2+b^2}{(a^2+b^2)^2}}$$

$$= \sqrt{\frac{1}{a^2+b^2}}$$

$$= \frac{1}{\sqrt{a^2+b^2}}$$

$$= \frac{1}{|z|}$$

(3)

$$\text{b) } (z-1)^4 + (z+1)^4 = 0$$

$$(z-1)^4 = -(z+1)^4$$

$$\left(\frac{z-1}{z+1}\right)^4 = -1 = \operatorname{cis}(\pi + 2k\pi)$$

$$\text{Let } y = \frac{z-1}{z+1} \Rightarrow y^4 = \operatorname{cis}(\pi + 2k\pi)$$

$$\therefore y = \operatorname{cis}\left(\frac{\pi + 2k\pi}{4}\right)$$

$$\text{let } \theta = \frac{\pi + 2k\pi}{4}$$

$$k = 0, 1, 2, 3$$

then

$$y = \frac{z-1}{z+1} = \cos \theta + i \sin \theta$$

$$z-1 = (z+1)(\cos \theta + i \sin \theta)$$

$$z-1 = z \cos \theta + i z \sin \theta + \cos \theta + i \sin \theta$$

$$z = z \cos \theta + i z \sin \theta + \cos \theta + i \sin \theta + 1$$

$$z - z \cos \theta - i z \sin \theta = \cos \theta + i \sin \theta + 1$$

$$z(1 - \cos \theta - i \sin \theta) = \cos \theta + i \sin \theta + 1$$

$$z = \frac{\cos \theta + i \sin \theta + 1}{1 - \cos \theta - i \sin \theta}$$

$$= \frac{2 \cos^2 \frac{\theta}{2} + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2} - 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

$$= \frac{2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})}{2 \sin \frac{\theta}{2} (\sin \frac{\theta}{2} - i \cos \frac{\theta}{2})} \times \frac{i}{i}$$

$$= i \cot \frac{\theta}{2} \left(\frac{\cos \frac{\theta}{2} + i \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} + i \sin \frac{\theta}{2}} \right)$$

$$= i \cot \frac{\theta}{2} ; \theta = \frac{\pi + 2k\pi}{4}$$

$$k = 0, 1, 2, 3$$

$$\therefore z_1 = i \cot \frac{\pi}{8}$$

$$z_2 = i \cot \frac{3\pi}{8}$$

$$z_3 = i \cot \frac{5\pi}{8} = i \cot \frac{-3\pi}{8}$$

$$z_4 = i \cot \frac{7\pi}{8} = i \cot \frac{-\pi}{8}$$

$$\text{Roots: } \pm i \cot \frac{\pi}{8}, \pm i \cot \frac{3\pi}{8} \quad (6)$$

